INCH-POUND

MIL-C-70461D (AR)
AMENDMENT 1
13 May 1992

MILITARY SPECIFICATION

COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS
FOR CHARGE, PROPELLING, 155MM, M203A1

This Amendment forms a part of Military Specification MIL-C-70461D (AR) dated 28 May 1991, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

PAGE 16

4.5.3.3.2.2 Delete in its entirely and substitute new paragraph 4.5.3.3.2.2:

"4.5.3.3.2.2 Method 2 (UV method). Cut 1.0 to 1.1 grams of sample into 1/4 inch squares or smaller. Dry in microwave moisture analyzer. Transfer the dried sample to a weighing bottle. Cool the sample to room temperature in a desiccate and weigh. Transfer the sample to a square homogenizing beaker. Reweigh the empty weighing bottle. Sample weight = total weight minus weight of empty weighing bottle. Add 50 ml of 90% methanol to the square beaker. Agitate the sample/solvent mixture in the hot ultrasonic bath for 10 minutes. Homogenize the mixture using a Binkmann homogenizer (or equivalent) for 2 minutes or until the sample is completely defiberized. Transfer the mixture to a filtering apparatus fitted with a suitable filter paper.

Add fresh 90% methanol to the square beaker. Grind this solution in the polytron for 1 minute. Add to the filtering apparatus. Repeat two times. Transfer the filtrate into a 250 ml volumetric flask. Make up to the mark with 90% methanol. Shake the flask well enough to make sure the solution is uniform. Using a pipet, transfer exactly 15 ml of this solution into a 50 ml volumetric flask. Make up to mark with 90% methanol. Shake well. Measure the absorbance of this solution at 285 nm. Calculate the DPA content according to the following formulas:

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$$\text{DPA \%} = \frac{(Y - a)\ 250}{b * w}$$

Where y = absorbance
a = intercept of calibration curve
b = slope of calibration curve
w = sample of weight in gms."

PAGE 17

Add the following new paragraph 4.5.3.3.2.2.1:

"4.5.3.3.2.2.1 Calibration procedure. Dry approximately 1 gram of diphenylamine using the microwave moisture analyzer.

Weight approximately 200 mg to the nearest 0.2 mg. Add this to a 250 ml volumetric flask. Add 90% methanol solvent and dilute to mark. Mix well.

Pipet 5 mls from the 250 ml flask and transfer to a 100 ml volumetric flask. Dilute to mark using the same solvent and mix well. This is the standard stock solution. Volumetrically pipet the following aliquot of the standard solution into 100 ml volumetric flasks.

5 ml solution in a 100 ml volumetric flask = 0.002 mg/ml
10 ml solution in a 100 ml volumetric flask = 0.004 mg/ml
15 ml solution in a 100 ml volumetric flask = 0.006 mg/ml

Dilute all standard solutions to mark and mix well. Zero the spectrophotometer with fresh 90% methanol solvent at 285 nm. Rinse the cell with the first standard twice, then measure the absorbance.

Rinse the same cell with the second standard, then measure its absorbance. Perform the same procedure with the 3rd standard. Input the absorbance and concentration data points into a linear regression program. Obtain the slope, y-intercept and r value (correlation coefficient). If r is less than 0.999, the standard should be rerun."

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